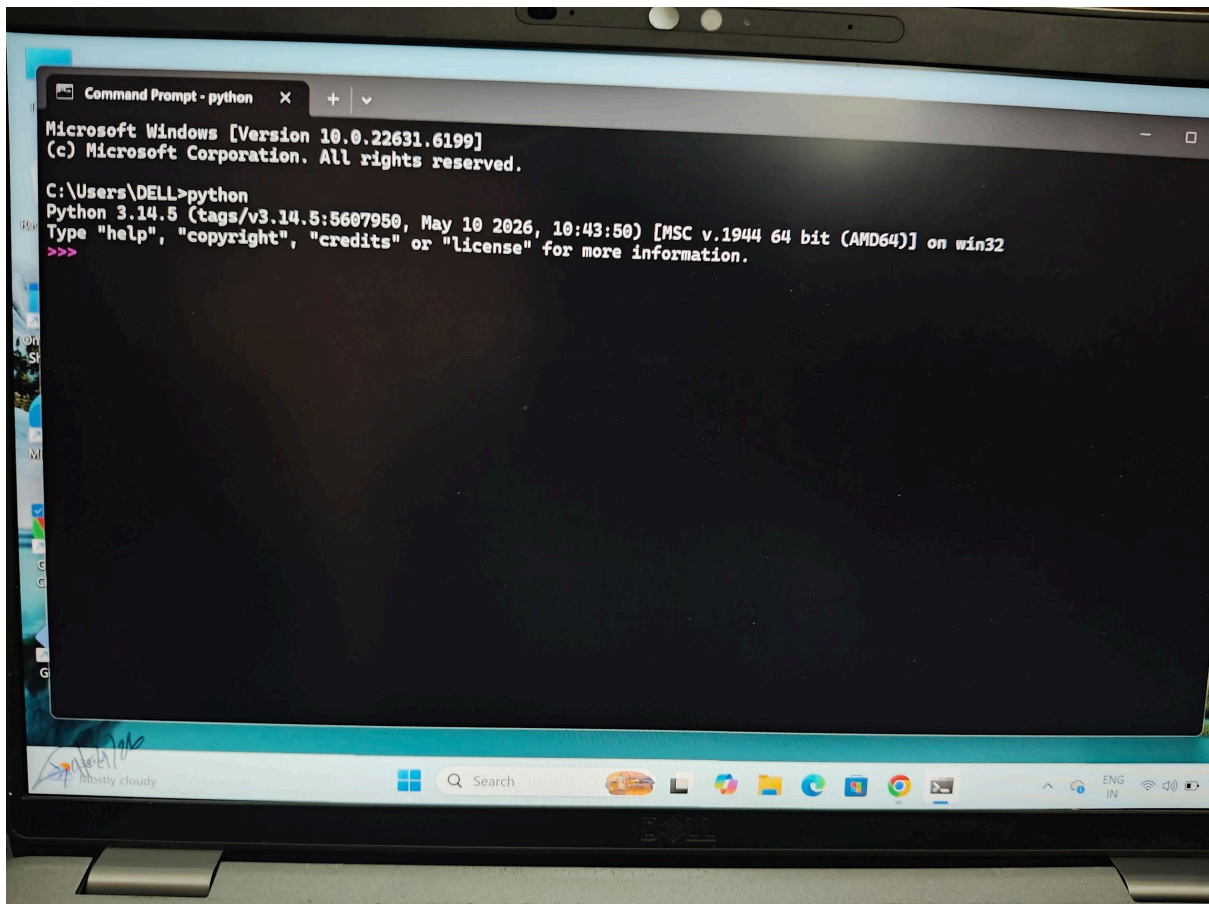


Tab 1

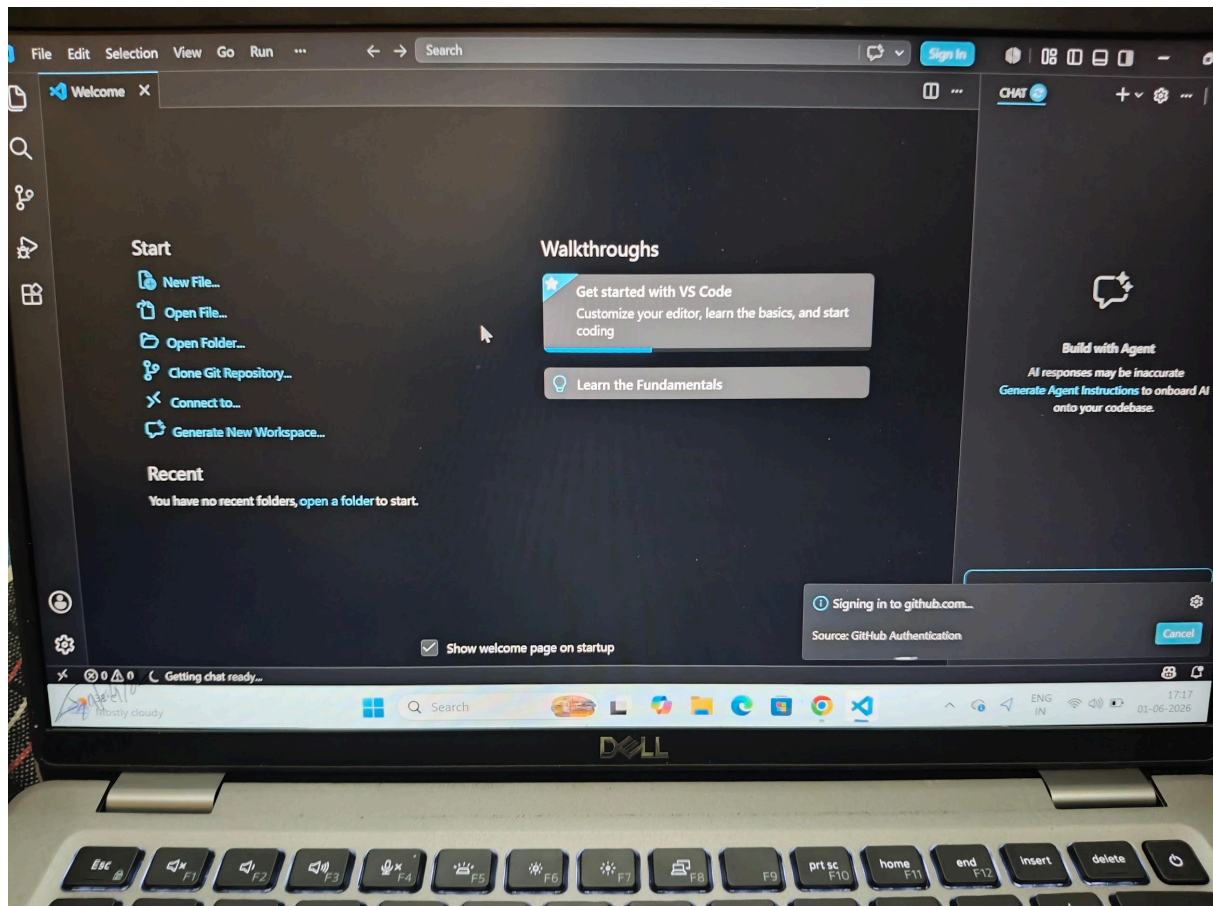
TASK TO COMPLETE

1 Install Required Tools

PYTHON

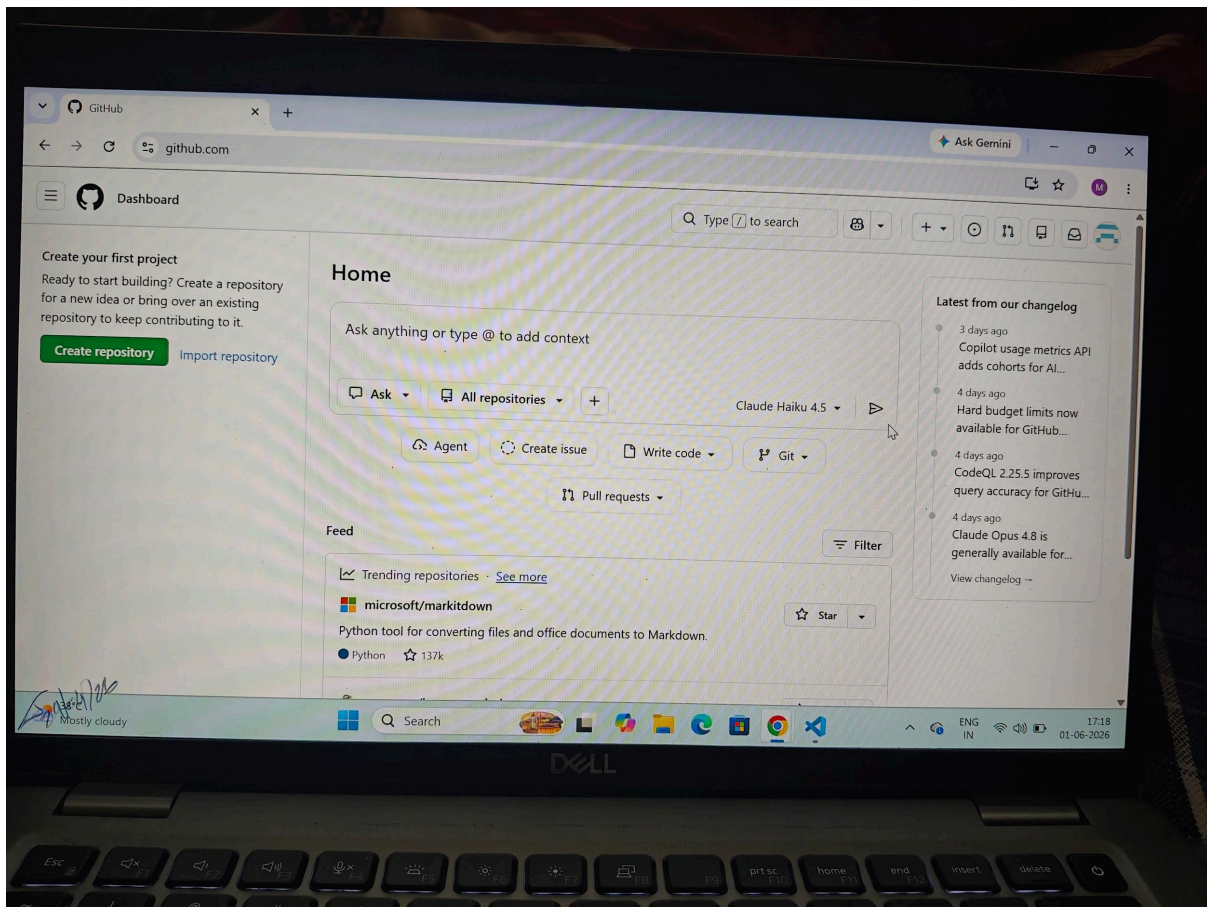


Vs code



GIT

2 GITHUB ACCOUNT



3. RESEARCH TOPICS

1. What is SLM (Small Language Model)?

SLM stands for Small Language Model. It is an AI model trained on text data like an LLM but has fewer parameters and requires less computing power.

Features:

- Faster response times
- Lower memory and storage requirements
- Can run on laptops, smartphones, and edge devices
- Less expensive to deploy

Examples:

- Phi-3
- Gemma
- Llama 3 8B

Uses:

- Chatbots
- Mobile AI applications
- Offline AI assistants

2. What is a RAG System?

RAG stands for Retrieval-Augmented Generation.

It is a technique that allows an AI model to retrieve relevant information from external sources (documents, databases, websites, PDFs, etc.) before generating an answer.

Why RAG?

Language models may not know the latest information or organization-specific data. RAG helps them access relevant information in real time.

Benefits:

More accurate answers

Access to up-to-date information

Reduced hallucinations

Useful for company knowledge bases and document search

3. Difference Between LLM and SLM

LLM (Large Language Model) and SLM (Small Language Model) are both types of artificial intelligence models used to understand and generate human language, but they differ in size and capability. LLMs contain a very large number of parameters, allowing them to handle complex tasks, advanced reasoning, and a wide range of topics with high accuracy.

However, they require more computing power, memory, and cost to operate. In contrast, SLMs are smaller and more efficient, using fewer resources and providing faster responses.

They are well-suited for simpler tasks, on-device applications, and situations where speed and low cost are important. While LLMs generally offer better performance and broader knowledge, SLMs provide a practical and efficient solution for specific or lightweight AI applications.

Examples

LLMs: GPT, Claude

SLMs: Phi-3, Gemma

4. How RAG Works with LLMs and SLMs

Step 1: User Asks a Question

Example:

"What are the attendance rules for our college?"

Step 2: Retrieval

The RAG system searches relevant documents, databases, or PDFs and finds useful information.

Step 3: Context Injection

The retrieved information is attached to the user's query.

Step 4: Generation

The LLM or SLM reads the retrieved information and generates an answer based on that data.

Workflow

User Question



Document Search



Retrieve Relevant Data



Send Data + Question to Model



LLM/SLM Generates Answer



Final Response

- PROBLEMS THAT I HAVE BEEN PLACED WAS IN INSTALLING AND CREATING GIT IN LAPTOP SO THEN I TOOK THE HELP OF CHATGPT I GOT THAT .
- I HAVE LEARNED VERY CLEARLY ABOUT INSTALLATION AND SET UPping ACCOUNT AND LEARNING ABOUT SLM ,RAG ,LLM AND PREPARING NOTES AND HOW TO DO DOCUMENTATION .